

Instructions:

- You may use any calculator except a TI-nspire.
- Put away all phones, laptops, and course materials.
- Show all of your work, and use correct notation. Failure to do these things will result in a deduction of points.
- You have **1 hour** to complete this exam.
- I'm proud of you—you've got this!

1. (8 pts.) With any hypothesis test, we could make the wrong decision. In the table below, determine the cells that correspond to a wrong decision, and write in these cells the type of error that is being made. Leave the other cells blank.

		Decision	
		Reject H_0	Fail to Reject H_0
Reality	H_0 is true		
	H_0 is false		

Solution:

		Decision	
		Reject H_0	Fail to Reject H_0
Reality	H_0 is true	Type I error	
	H_0 is false		Type II error

2. (12 pts.) Let $Y_1, \dots, Y_n$ be independent random variables, each with probability density function

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}.$$

Show that $\bar{Y}$ converges in probability to $2/3$.

Solution:

$$E(\bar{Y}) = E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = E(Y_i) = \int_0^1 2y^2 dy = \frac{2}{3} y^3 \Big|_0^1 = \frac{2}{3}.$$

$$V(\bar{Y}) = V\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n^2} \sum_{i=1}^n V(Y_i) = \frac{1}{n} V(Y_i) = \frac{1}{18n},$$

where $V(Y_i)$ is calculated as follows:

$$\begin{aligned} V(Y_i) &= E(Y_i^2) - [E(Y_i)]^2 \\ &= \int_0^1 2y^3 dy - \frac{4}{9} \\ &= \left. \frac{y^4}{2} \right|_0^1 - \frac{4}{9} \\ &= \frac{1}{2} - \frac{4}{9} \\ &= \frac{1}{18}. \end{aligned}$$

Clearly, $\lim_{n \rightarrow \infty} V(\bar{Y}) = 0$. Hence, by Theorem 9.1, $\bar{Y}$ converges in probability to $2/3$.

3. (16 pts.) Let $Y_1, \dots, Y_n$ denote a random sample from a population whose probability density function is given by

$$f(y|\theta) = \begin{cases} \frac{3}{\theta} y^2 e^{-y^3/\theta}, & y > 0 \\ 0, & \text{otherwise} \end{cases}.$$

- (a.) (12 pts.) Use the Factorization Criterion to show that $\sum_{i=1}^n Y_i^3$ is a sufficient statistic for θ .

Solution:

$$\begin{aligned} L(\theta) &= f(y_1|\theta) \cdot f(y_2|\theta) \cdot \dots \cdot f(y_n|\theta) \\ &= \frac{3}{\theta} y_1^2 e^{-y_1^3/\theta} \cdot \frac{3}{\theta} y_2^2 e^{-y_2^3/\theta} \cdot \dots \cdot \frac{3}{\theta} y_n^2 e^{-y_n^3/\theta} \\ &= \left(\frac{3}{\theta}\right)^n \cdot \prod_{i=1}^n y_i^2 \cdot e^{-\frac{1}{\theta} \sum_{i=1}^n y_i^3} \\ &= \underbrace{\left(\frac{3}{\theta}\right)^n e^{-\frac{1}{\theta} \sum_{i=1}^n y_i^3}}_{g\left(\sum_{i=1}^n y_i^3, \theta\right)} \cdot \underbrace{\prod_{i=1}^n y_i^2}_{h(y_1, \dots, y_n)} \end{aligned}$$

Hence, by the Factorization Criterion, $\sum_{i=1}^n Y_i^3$ is a sufficient statistic for θ .

(b.) (4 pts.) Is this sufficient statistic unique? Why or why not?

Solution: No! Any one-to-one function of a sufficient statistic is sufficient as well. So, for example, $\frac{1}{n} \sum_{i=1}^n Y_i^3$ would also be a sufficient statistic for θ .

4. (28 pts.) Let $Y_1, \dots, Y_n$ constitute a random sample from the probability density function given by

$$f(y|\theta) = \begin{cases} \theta y^{\theta-1}, & 0 < y < 1, \quad \theta > 0 \\ 0, & \text{otherwise} \end{cases}.$$

- (a.) (12 pts.) Find the method of moments estimator, $\hat{\theta}_{MOM}$, for θ .

Solution:

$$\mu'_1 = E(Y) = \int_0^1 y \cdot \theta y^{\theta-1} dy = \int_0^1 \theta y^\theta dy = \frac{\theta}{\theta+1} y^{\theta+1} \Big|_0^1 = \frac{\theta}{\theta+1}.$$

$$m'_1 = \frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y}.$$

Setting the first population and sample moments equal to one another yields the following:

$$\begin{aligned} \bar{Y} = \frac{\theta}{\theta+1} &\implies (\theta+1)\bar{Y} = \theta \\ &\implies \theta\bar{Y} - \theta = -\bar{Y} \\ &\implies \theta(\bar{Y} - 1) = -\bar{Y} \\ &\implies \theta = \frac{\bar{Y}}{1 - \bar{Y}}. \end{aligned}$$

Hence, $\hat{\theta}_{MOM} = \bar{Y}/(1 - \bar{Y})$.

(b.) (12 pts.) Find the maximum likelihood estimator, $\hat{\theta}_{MLE}$, for θ .

Solution:

$$\begin{aligned} L(\theta) &= f(y_1|\theta) \cdot f(y_2|\theta) \cdot \dots \cdot f(y_n|\theta) \\ &= \theta y_1^{\theta-1} \cdot \theta y_2^{\theta-1} \cdot \dots \cdot \theta y_n^{\theta-1} \\ &= \theta^n \prod_{i=1}^n y_i^{\theta-1} \end{aligned}$$

$$\begin{aligned} \ln L(\theta) &= n \cdot \ln \theta + \sum_{i=1}^n \ln(y_i^{\theta-1}) \\ &= n \cdot \ln \theta + (\theta - 1) \sum_{i=1}^n \ln(y_i) \end{aligned}$$

$$\frac{d[\ln L(\theta)]}{d\theta} = \frac{n}{\theta} + \sum_{i=1}^n \ln(y_i) \stackrel{\text{set } = 0}{\implies} \frac{n}{\hat{\theta}_{MLE}} = - \sum_{i=1}^n \ln(y_i) \implies \hat{\theta}_{MLE} = - \frac{n}{\sum_{i=1}^n \ln(y_i)}$$

$$\frac{d^2[\ln L(\theta)]}{d\theta^2} = -\frac{n}{\theta^2} < 0$$

Hence, $\hat{\theta}_{MLE} = -n / \sum_{i=1}^n \ln(y_i)$.

(c.) (4 pts.) What is the MLE of $e^{-\theta}$? Justify your answer.

Solution: The Invariance Property of MLEs implies that the MLE of $e^{-\theta}$ is $e^{-\hat{\theta}_{MLE}} = e^{n / \sum_{i=1}^n \ln(y_i)}$.

5. (24 pts.) Let $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Poisson}(\lambda)$.

(a.) (12 pts.) Find a minimal sufficient statistic (MSS) for λ .

Solution: The likelihood function can be calculated as follows:

$$\begin{aligned} L(y_1, \dots, y_n | \lambda) &= f(y_1 | \lambda) \cdot f(y_2 | \lambda) \cdot \dots \cdot f(y_n | \lambda) \\ &= \frac{\lambda^{y_1} e^{-\lambda}}{y_1!} \cdot \frac{\lambda^{y_2} e^{-\lambda}}{y_2!} \cdot \dots \cdot \frac{\lambda^{y_n} e^{-\lambda}}{y_n!} \\ &= \frac{\lambda^{\sum_{i=1}^n y_i} e^{-n\lambda}}{\prod_{i=1}^n y_i!}. \end{aligned}$$

We'll now use the Lehmann-Scheffé Technique. First, we need to calculate R :

$$R = \frac{L(x_1, \dots, x_n | \lambda)}{L(y_1, \dots, y_n | \lambda)} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} \cdot \frac{\prod_{i=1}^n y_i!}{e^{-n\lambda} \lambda^{\sum_{i=1}^n y_i}} = \frac{\prod_{i=1}^n y_i!}{\prod_{i=1}^n x_i!} \cdot \lambda^{\sum_{i=1}^n x_i - \sum_{i=1}^n y_i}.$$

Second, we note that R is free of λ if and only if $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$.

Hence, $U = \sum_{i=1}^n Y_i$ is a MSS for λ .

- (b.) (12 pts.) Find a minimum variance unbiased estimator (MVUE) for λ . Justify all of your work.

Solution: If we can find an unbiased estimator that is a function of the MSS derived in part (a.), this estimator must be a MVUE.

$$E(U) = E\left(\sum_{i=1}^n Y_i\right) = n\lambda.$$

Define $\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y}$. Then,

$$E(\hat{\lambda}) = E(\bar{Y}) = \frac{1}{n} \sum_{i=1}^n E(Y_i) = \frac{1}{n} \cdot n\lambda = \lambda.$$

Since $\hat{\lambda} = \bar{Y}$ is both an unbiased estimator of λ and a function of the previously derived MSS, $\hat{\lambda}$ must be a MVUE.

Choose **one** of problems 6-8 to complete. If you work on more than one problem, indicate which one you want graded; otherwise, your instructor may randomly select which one to grade.

6. (12 pts.) Prove the following theorem:

Theorem 9.5: Rao-Blackwell Theorem. Let $\hat{\theta}$ be an unbiased estimator for θ such that $V(\hat{\theta}) < \infty$. If U is a sufficient statistic for θ , define $\hat{\theta}^* = E(\hat{\theta}|U)$. Then,

$$E(\hat{\theta}^*) = \theta \quad \text{and} \quad V(\hat{\theta}^*) \leq V(\hat{\theta}).$$

Solution: First, we'll show that $E(\hat{\theta}^*) = \theta$:

$$E(\hat{\theta}^*) = E[E(\hat{\theta}|U)] = E(\hat{\theta}) = \theta,$$

where the penultimate equality is due to the Law of Iterated Expectations and the last equality holds because $\hat{\theta}$ is unbiased for θ .

Using the Law of Total Variance, we know that

$$\begin{aligned} V(\hat{\theta}) &= E[V(\hat{\theta}|U)] + V[E(\hat{\theta}|U)] \\ &= E[V(\hat{\theta}|U)] + V(\hat{\theta}^*). \end{aligned}$$

Note that $V(\hat{\theta}|U = u) \geq 0$ for all u . Thus, $E[V(\hat{\theta}|U)] \geq 0$.

So,

$$V(\hat{\theta}^*) = V(\hat{\theta}) - \underbrace{E[V(\hat{\theta}|U)]}_{\geq 0} \implies V(\hat{\theta}^*) \leq V(\hat{\theta}).$$

Hence, the theorem is proved.

7. (12 pts.) Suppose $Y_1, \dots, Y_n$ are i.i.d. such that $E(Y_i) = \mu$, $V(Y_i) = \sigma^2$, $E(Y_i^2) = \mu'_2$, and $E(Y_i^4) = \mu'_4$ are all finite. Prove that $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y}_n)^2 \xrightarrow{p} \sigma^2$.

NOTE: You may use the fact that S_n^2 can be rewritten as

$$S_n^2 = \frac{n}{n-1} \left[\frac{1}{n} \sum_{i=1}^n Y_i^2 - \bar{Y}_n^2 \right].$$

Solution: Let's first look at the following 3 components separately: $\frac{n}{n-1}$, $\frac{1}{n} \sum_{i=1}^n Y_i^2$, and $\bar{Y}_n^2$.

$$\frac{n}{n-1} : \lim_{n \rightarrow \infty} \frac{n}{n-1} = 1, \text{ so } \frac{n}{n-1} \xrightarrow{p} 1.$$

$\frac{1}{n} \sum_{i=1}^n Y_i^2$: $Y_1^2, \dots, Y_n^2$ are an i.i.d. sample from a distribution with mean $E(Y_i^2) = \mu'_2$ and $V(Y_i^2) = E(Y_i^4) - [E(Y_i^2)]^2 = \mu'_4 - (\mu'_2)^2 < \infty$. Then, by the Law of Large Numbers, $\frac{1}{n} \sum_{i=1}^n Y_i^2 \xrightarrow{p} E(Y_i^2) = \mu'_2$.

$\bar{Y}_n^2$: By the Law of Large Numbers, $\bar{Y}_n \xrightarrow{p} \mu$. Thus, by Theorem 9.2(e.), $\bar{Y}_n^2 \xrightarrow{p} \mu^2$.

Therefore, by Theorem 9.2(b.), $\frac{1}{n} \sum_{i=1}^n Y_i^2 - \bar{Y}_n^2 \xrightarrow{p} \mu'_2 - \mu^2 = E(Y_i^2) - [E(Y_i)]^2 = V(Y_i)$.

Then, by Theorem 9.2(c.), $S_n^2 = \frac{n}{n-1} \left[\frac{1}{n} \sum_{i=1}^n Y_i^2 - \bar{Y}_n^2 \right] \xrightarrow{p} V(Y_i) = \sigma^2$.

8. (12 pts.) Prove the following theorem:

Theorem 9.1: If $\hat{\theta}_n$ is an unbiased estimator of θ and $\lim_{n \rightarrow \infty} V(\hat{\theta}_n) = 0$, then $\hat{\theta}_n$ is a consistent estimator of θ .

Solution: Suppose $E(\hat{\theta}_n) = \theta$ and $\lim_{n \rightarrow \infty} V(\hat{\theta}_n) = 0$. Let $\epsilon > 0$. Define $k = \epsilon/\sigma_{\hat{\theta}_n} > 0$.

WTS: $\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$.

$$\begin{aligned} P(|\hat{\theta}_n - \theta| > \epsilon) &\leq P(|\hat{\theta}_n - \theta| \geq \epsilon) \\ &= P(|\hat{\theta}_n - \theta| \geq k\sigma_{\hat{\theta}_n}) \\ &\leq \frac{1}{k^2} \quad (\text{Tchebysheff's Theorem}) \\ &= \frac{\sigma_{\hat{\theta}_n}^2}{\epsilon^2} \\ &= \frac{V(\hat{\theta}_n)}{\epsilon^2}. \end{aligned}$$

Thus, we have the following:

$$\begin{aligned} 0 \leq P(|\hat{\theta}_n - \theta| > \epsilon) &\leq \frac{V(\hat{\theta}_n)}{\epsilon^2} \implies \lim_{n \rightarrow \infty} 0 \leq \lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) \leq \frac{1}{\epsilon^2} \underbrace{\lim_{n \rightarrow \infty} V(\hat{\theta}_n)}_{=0} \\ &\implies 0 \leq \lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) \leq 0. \end{aligned}$$

Hence, by the Squeeze Theorem, $\lim_{n \rightarrow \infty} P(|\hat{\theta}_n - \theta| > \epsilon) = 0$. Therefore, $\hat{\theta}_n$ is a consistent estimator of θ .